Partial Differential Equations

for Scientists and Engineers

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حمیدرضا یاوری پور

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Preface

Over the last few years, there has been a significant increase in the number of students studying partial differential equations at the undergraduate level, and many of these students have come from areas other than mathematics, where intuition rather than mathematical rigor is emphasized. In writing *Partial Differential Equations for Scientists and Engineers*, I have tried to stimulate intuitive thinking, while, at the same time, not losing too much mathematical accuracy. At one extreme, it is possible to approach the subject on a high mathematical epsilon-delta level, which generally results in many undergraduate students not knowing what's going on. At the other extreme, it is possible to wave away all the subtleties until neither the student *nor the teacher knows what's going on.* I have tried to steer the mathematical thinking somewhere between these two extremes.

Partial Differential Equations for Scientists and Engineers evolved from a set of lecture notes I have been preparing for the last five years. It is an unconventional text in one regard: It is organized in 47 semi-independent lessons in contrast to the more usual chapter-by-chapter approach.

Separation of variables and integral transforms are the two most important analytic tools discussed. Several nonstandard topics, such as Monte Carlo methods, calculus of variations, control theory, potential theory, and integral equations, are also discussed because most students will eventually come across these subjects at some time in their studies. Unless they study these topics here, they will probably never study them formally.

This book can be used for a one- or two-semester course at the junior or senior level. It assumes only a knowledge of differential and integral calculus and ordinary differential equations. Most lessons take either one or two days, so that a typical one-semester syllabus would be: Lessons 1–13, 15–17, 19–20, 22–23, 25–27, 30–32, 37–39. All 47 lessons can easily be covered in two semesters, with plenty of time to work problems.

The author wishes to thank the editors at Wiley for their invitation to write this book as well as the reviewers, Professor Chris Rorres and Professor M. Kursheed Ali, who helped me greatly with their suggestions. Any further suggestions for improvement of this book, either from students or teachers, would be greatly appreciated. Thanks also to Dorothy, Susan, Alexander, and Daisy Farlow.

Stanley J. Farlow

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Partial Differential Equations

for
Scientists and
Engineers

PART 1

Introduction

LESSON 1

Introduction to Partial Differential Equations

PURPOSE OF LESSON: To show what partial differential equations are, why they are useful, and how they are solved; also included is a brief discussion on how they are classified as various kinds and types. An overview is given of many of the ideas that will be studied in detail later.

Most physical phenomena, whether in the domain of fluid dynamics, electricity, magnetism, mechanics, optics, or heat flow, can be described in general by partial differential equations (PDEs); in fact, most of mathematical physics are PDEs. It's true that simplifications can be made that reduce the equations in question to ordinary differential equations, but, nevertheless, the complete description of these systems resides in the general area of PDEs.

What Are PDEs?

A partial differential equation is an equation that contains partial derivatives. In contrast to ordinary differential equations (ODEs), where the unknown function depends only on *one variable*, in PDEs, the unknown function depends on several variables (like temperature $u(x,t)$ depends both on location x and time t).

Let's list some well-known PDEs; note that for notational simplicity we have called

$$u_t = \frac{\partial u}{\partial t} \qquad u_x = \frac{\partial u}{\partial x} \qquad u_{xx} = \frac{\partial^2 u}{\partial x^2} \qquad \ldots$$

A Few Well-Known PDEs

$u_t = u_{xx}$ (heat equation in one dimension)

$u_t = u_{xx} + u_{yy}$ (heat equation in two dimensions)

$u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta} = 0$ (Laplace's equation in polar coordinates)

$$u_{tt} = u_{xx} + u_{yy} + u_{zz} \quad \text{(wave equation in three dimensions)}$$
$$u_{tt} = u_{xx} + \alpha u_t + \beta u \quad \text{(telegraph equation)}$$

Note on the Examples

The unknown function u always depends on *more* than one variable. The variable u (which we differentiate) is called the **dependent** variable, whereas the ones we differentiate *with respect to* are called the **independent** variables. For example, it is clear from the equation

$$u_t = u_{xx}$$

that the dependent variable $u(x,t)$ is a function of two independent variables x and t, whereas in the equation

$$u_t = u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta}$$

$u(r,\theta,t)$ depends on r, θ, and t.

Why Are PDEs Useful?

Most of the **natural laws of physics**, such as Maxwell's equations, Newton's law of cooling, the Navier-Stokes equations, Newton's equations of motion, and Schrodinger's equation of quantum mechanics, are stated (or can be) in terms of PDEs, that is, these laws describe physical phenomena by relating **space and time derivatives.** Derivatives occur in these equations because the derivatives represent *natural things* (like velocity, acceleration, force, friction, flux, current). Hence, we have equations relating partial derivatives of some unknown quantity that we would like to find.

The purpose of this book is to show the reader two things

1. How to *formulate* the PDE from the physical problem (constructing the mathematical model).
2. How to *solve* the PDE (along with initial and boundary conditions).

We wait a few lessons before we start the modeling problem; now, a brief overview on how PDEs are solved.

How Do You Solve a Partial Differential Equation?

This is a good question. It turns out that there is an entire arsenal of methods available to the practitioner: the most important methods are those that change PDEs into ODEs. Ten useful techniques are

1. *Separation of Variables.* This technique reduces a PDE in n variables to n ODEs.
2. *Integral Transforms.* This procedure reduces a PDE in n independent variables to one in $n - 1$ variables; hence, a PDE in two variables could be changed to an ODE.
3. *Change of Coordinates.* This method changes the original PDE to an ODE or else another PDE (an easier one) by changing the coordinates of the problem (rotating the axis and things like that).
4. *Transformation of the Dependent Variable.* This method transforms the unknown of a PDE into a new unknown that is easier to find.
5. *Numerical Methods.* These methods change a PDE to a system of *difference equations* that can be solved by means of iterative techniques on a computer; in many cases, this is the only technique that will work. In addition to methods that replace PDEs by difference equations, there are other methods that attempt to approximate solutions by polynomial surfaces (spline approximations).
6. *Pertubation Methods.* This method changes a nonlinear problem into a sequence of *linear ones* that approximates the nonlinear one.
7. *Impulse-response Technique.* This procedure decomposes initial and boundary conditions of the problem into *simple impulses* and finds the response to each impulse. The overall response is then found by adding these simple responses.
8. *Integral Equations.* This technique changes a PDE to an **integral equation** (an equation where the unknown is inside the integral). The integral equation is then solved by various techniques.
9. *Calculus of Variations Methods.* These methods find the solution to PDEs by reformulating the equation as a *minimization problem.* It turns out that the minimum of a certain expression (very likely the expression will stand for total energy) is also the solution to the PDE.
10. *Eigenfunction Expansion.* This method attempts to find the solution of a PDE as an infinite sum of *eigenfunctions*. These eigenfunctions are found by solving what is known as an eigenvalue problem corresponding to the original problem.

Kinds of PDEs

Partial differential equations are classified according to many things. Classification is an important concept because the general theory and methods of solution usually apply only to a given class of equations. Six basic classifications are

1. *Order of the PDE.* The order of a PDE is the order of the *highest partial derivative* in the equation, for example,

$$u_t = u_{xx} \quad \text{(second order)}$$

$$u_t = u_x \qquad \text{(first order)}$$
$$u_t = uu_{xxx} + \sin x \qquad \text{(third order)}$$

2. *Number of Variables.* The number of variables is the *number of independent* variables, for example,

$$u_t = u_{xx} \qquad \text{(two variables: } x \text{ and } t\text{)}$$
$$u_t = u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta} \qquad \text{(three variables: } r, \theta, \text{ and } t\text{)}$$

3. *Linearity.* Partial differential equations are either *linear* or *nonlinear*. In the linear ones, the dependent variable u and all its derivatives appear in a linear fashion (they are not multiplied together or squared, for example). More precisely, a **second-order linear equation in two variables** is an equation of the form

$$\boxed{Au_{xx} + Bu_{xy} + Cu_{yy} + Du_x + Eu_y + Fu = G} \tag{1.1}$$

where A, B, C, D, E, F, and G can be *constants* or given *functions* of x and y; for example,

$$u_{tt} = e^{-t}u_{xx} + \sin t \qquad \text{(linear)}$$
$$uu_{xx} + u_t = 0 \qquad \text{(nonlinear)}$$
$$u_{xx} + yu_{yy} = 0 \qquad \text{(linear)}$$
$$xu_x + yu_y + u^2 = 0 \qquad \text{(nonlinear)}$$

4. *Homogeneity.* The equation (1.1) is called **homogeneous** if the right-hand side $G(x,y)$ is identically zero for all x and y. If $G(x,y)$ is not identically zero, then the equation is called **nonhomogeneous**.
5. *Kinds of Coefficients.* If the coefficients A, B, C, D, E, and F in equation (1.1) are constants, then (1.1) is said to have **constant coefficients** (otherwise, variable coefficients).
6. *Three Basic Types of Linear Equations.* All linear PDEs like equation (1.1) are either
   (a) parabolic
   (b) hyperbolic
   (c) elliptic

Parabolic. **Parabolic equations** describe heat flow and diffusion processes and satisfy the property $B^2 - 4AC = 0$.

Hyperbolic. **Hyperbolic equations** describe vibrating systems and wave motion and satisfy the property $B^2 - 4AC > 0$.

Elliptic. **Elliptic equations** describe *steady-state* phenomena and satisfy the property $B^2 - 4AC < 0$.

Examples.

(a) $u_t = u_{xx}$ $\quad B^2 - 4AC = 0$ $\quad$ (parabolic)

(b) $u_{tt} = u_{xx}$ $\quad B^2 - 4AC = 4$ $\quad$ (hyperbolic)

(c) $u_{\xi\eta} = 0$ $\quad B^2 - 4AC = 1$ $\quad$ (hyperbolic)

(d) $u_{xx} + u_{yy} = 0$ $\quad B^2 - 4AC = -4$ $\quad$ (elliptic)

(e) $yu_{xx} + u_{yy} = 0$ $\quad B^2 - 4AC = -4y$ $\quad \begin{cases} \text{elliptic for } y > 0 \\ \text{parabolic for } y = 0 \\ \text{hyperbolic for } y < 0 \end{cases}$

(In the case of variable coefficients, the situation can change from point to point.)

NOTES

1. In general, $B^2 - 4AC$ is a *function* of the independent variables; hence, an equation can change from one basic type to another throughout the domain of the equation (although it's not common).
2. The general linear equation (1.1) was written with independent variables x and y. In many problems, one of the two variables stands for time and hence would be written in terms of x and t.
3. A *general classification* diagram is given in Figure 1.1.

Linearity	Linear				Nonlinear		
Order	1	2	3	4	5	...	m
Kinds of coefficients (linear equations)	Constant				Variable		
Homogeneity (linear equations)	Homogeneous				Nonhomogeneous		
Number of variables	1	2	3	4	5	...	n
Basic type (linear equations)	Hyperbolic			Parabolic		Elliptic	

FIGURE 1.1 Classification diagram for partial differential equations.

PROBLEMS

1. Classify the following equations according to all the properties we've discussed in Figure 1.1:

   (a) $u_t = u_{xx} + 2u_x + u$

   (b) $u_t = u_{xx} + e^{-t}$

   (c) $u_{xx} + 3u_{xy} + u_{yy} = \sin x$

   (d) $u_{tt} = uu_{xxxx} + e^{-t}$

2. How many solutions to the PDE $u_t = u_{xx}$ can you find? Try solutions of the form $u(x,t) = e^{ax+bt}$.
3. If $u_1(x,y)$ and $u_2(x,y)$ satisfy equation (1.1), then is it true that the sum satisfies it?; if yes, prove it.
4. Probably the easiest of all PDEs to solve is the equation

$$\frac{\partial u(x,y)}{\partial x} = 0$$

   Can you solve this equation? (Find all functions $u(x,y)$ that satisfy it.)
5. What about the PDE

$$\frac{\partial^2 u(x,y)}{\partial x \partial y} = 0$$

   Can you find all solutions $u(x,y)$ to this equation? (How many are there?) How does this compare with an ODE like

$$\frac{d^2y}{dx^2} = 0$$

   insofar as the number of solutions is concerned?

OTHER READING

1. *Elementary Partial Differential Equations* by P. W. Berg and J. L. McGregor. Holden-Day, 1966. Clearly written with several nice problems; a nice book to own.

2. *Analysis and Solution of Partial Differential Equations* by R. L. Street. Brooks-Cole, 1973. A well-written text covering many of the topics we will cover in this book.

PART 2

Diffusion-Type Problems

LESSON 2

Diffusion-Type Problems (Parabolic Equations)

PURPOSE OF LESSON: To show how parabolic PDEs are used to model heat-flow and diffusion-type problems. The physical meaning of different terms (such as u_t, u_x, u_{xx}, and u) are explained and a few examples of parabolic equations presented.

The idea of an *initial-boundary-value problem* is introduced along with an example. One of the major goals of this lesson is to give the reader an intuitive feeling for parabolic-type problems.

We begin this lesson by introducing a simple physical problem and showing how it can be described by means of a mathematical model (which will involve a PDE). We then complicate the problem and show how new partial differential equations can describe the new situations. The partial differential equations in this lesson are not derived or solved now, but will be in later lessons.

A Simple Heat-Flow Experiment

Suppose we have the following simple experiment that we break into steps:

STEP 1 We start with a reasonably long (say $L = 2$ m) rod (say copper) 2 cm in diameter whose lateral sides (but not the ends) we wrap with insulation. We could even use copper tubing provided we pour some sort of insulation down the inside. In other words, heat can flow in and out of the rod *at the ends*, but not across the lateral boundary.

STEP 2 Next, we place this rod in an environment whose temperature is fixed at some temperature T_0 (degrees °C) for a sufficiently long time, so that the temperature of the entire rod comes to a steady-state temperature similar to the environment. For simplicity, we let the temperature of the environment $T_0 = 10$°C.

STEP 3 We take the rod out of the environment at a time that we call $t = 0$ and attach two *temperature elements* to the ends of the rod. The purpose of these elements is to keep the ends at specific temperatures T_1 and T_2 (say $T_1 = 0°C$ and $T_2 = 50°C$). In other words, two thermostats constantly monitor the temperature at the ends of the rod, and if the temperatures differ from their prescribed values T_1 and T_2, strong heating (or cooling) elements come into operation to adjust the temperature accordingly. Our experiment is illustrated in Figure 2.1.

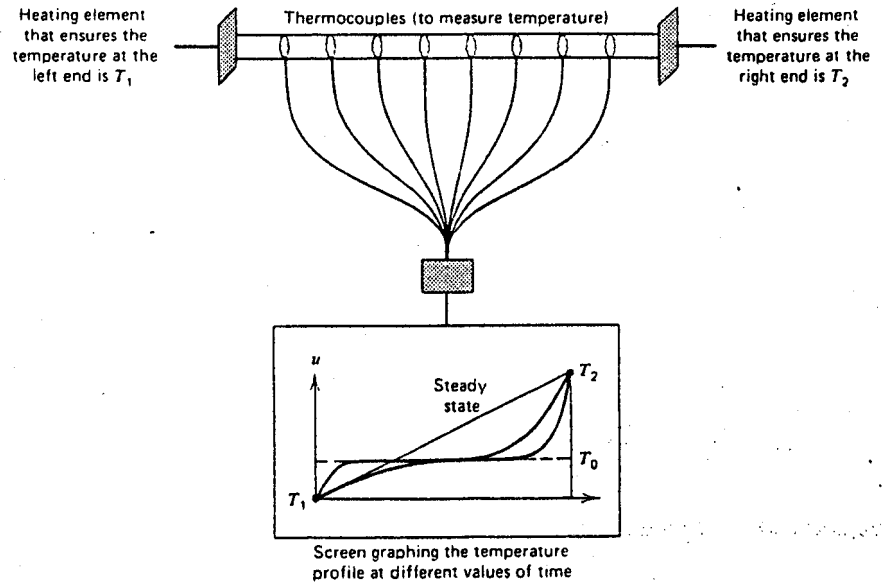

FIGURE 2.1 Schematic diagram of the experiment.

STEP 4 We now monitor the temperature profile of the rod on some type of display. (Why we want to perform an experiment of this kind is another question; we will talk about that later.) This completes our discussion of the experiment. The main purpose of this lesson is to show how this physical problem (and variations of it) can be explained (modeled) by parabolic PDEs.

The Mathematical Model of the Heat-Flow Experiment

The description of our physical problem requires three types of equations

1. The *PDE* describing the physical phenomenon of heat flow.
2. The *boundary conditions* describing the physical nature of our problem on the boundaries.
3. The *initial conditions* describing the physical phenomenon at the start of the experiment.

The Heat Equation

The basic equation of *one-dimensional* heat flow is the relationship

$$\text{(2.1)} \qquad \boxed{\text{PDE} \quad u_t = \alpha^2 u_{xx} \quad 0 < x < L \quad 0 < t < \infty}$$

which relates the quantities

u_t = the *rate of change* in temperature with respect to time (measured in deg/sec)

and

u_{xx} = the *concavity* of the temperature profile $u(x,t)$ (which essentially compares the temperature at one point to the temperature at neighboring points).

This equation will be derived from the basic *conservation of heat equation* in later lessons, but for the time being, we examine it by itself. This equation simply says that the temperature $u(x,t)$ (at some point along the rod x and at some point in time t) is increasing ($u_t > 0$) or decreasing ($u_t < 0$) according to whether u_{xx} is positive or negative. Figure 2.2 illustrates the change in temperature at different points along the rod.

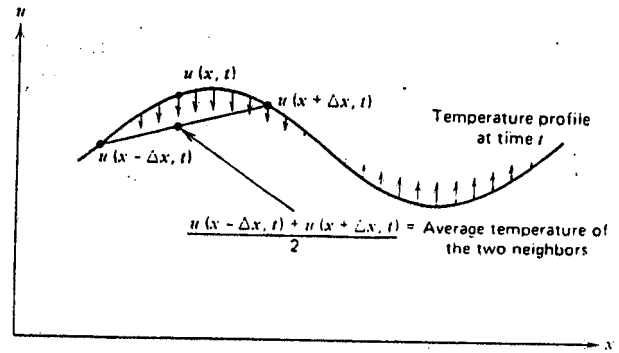

FIGURE 2.2 Arrows indicating change in temperature according to $u_t = \alpha^2 u_{xx}$

To see how u_{xx} can be interpreted to measure heat flow, suppose we approximate u_{xx} by the difference quotient

$$u_{xx}(x,t) \cong \frac{1}{\Delta x^2}[u(x + \Delta x,t) - 2u(x,t) + u(x - \Delta x,t)]$$

Since this can be rewritten

$$u_{xx}(x,t) \cong \frac{2}{\Delta x^2}\left[\frac{u(x+\Delta x,t)+u(x-\Delta x,t)}{2} - u(x,t)\right]$$

we have the following interpretation of u_{xx}:

1. If the temperature $u(x,t)$ < average of the two neighboring temperatures, then $u_{xx} > 0$ (here, the net flow of heat into x is positive).
2. If the temperature $u(x,t)$ = average of the two neighboring temperatures, then $u_{xx} = 0$ (here the net flow of heat into x is zero).
3. If the temperature $u(x,t)$ > average of the two neighboring temperatures, then $u_{xx} < 0$ (here the net flow of heat into x is negative).

This is illustrated in Figure 2.2. In other words, if the temperature at a point x is greater than the average of the temperature at two nearby points $x - \Delta x$ and $x + \Delta x$, then the temperature at x will be decreasing. Furthermore, the exact rate of decrease u_t is proportional to this difference. The proportionality constant α^2 is a property of the material, and we will discuss this constant more in the next few lessons.

Boundary Conditions

All physical problems have boundaries of some kind, so we must describe mathematically what goes on there in order to adequately describe the problem. In our experiment, the boundary conditions (BCs) are quite easy. Since the temperature u was fixed for all time $t > 0$ at T_1 and T_2 at the two ends $x = 0$ and $x = L$, we would simply say

$$\text{(2.2)} \qquad \text{BCs} \begin{cases} u(0,t) = T_1 \\ u(L,t) = T_2 \end{cases} \qquad 0 < t < \infty$$

Initial Conditions

All physical problems must start from some value of time (generally called $t = 0$), so we must specify the physical apparatus at this time. Since we started monitoring the rod temperature in our example from the time the rod had achieved a constant temperature of T_0, we have

$$\text{(2.3)} \qquad \boxed{\text{IC} \qquad u(x,0) = T_0 \qquad 0 \le x \le L}$$

We have now mathematically described the experiment. By writing equations (2.1), (2.2), and (2.3) together, we have what is called an *initial-boundary-value problem* (IBVP)

$$\text{(2.4)} \qquad \begin{array}{lll} \text{PDE} & u_t = \alpha^2 u_{xx} & 0 < x < L \quad 0 < t < \infty \\ \text{BCs} & \begin{cases} u(0,t) = T_1 \\ u(L,t) = T_2 \end{cases} & 0 < t < \infty \\ \text{IC} & u(x,0) = T_0 & 0 \le x \le L \end{array}$$

The interesting thing here, which is not at all obvious, is that there is *only one function* $u(x,t)$ that satisfies the problem (2.4), and that function will describe the temperature of the rod. Hence, our goal in the near future will be to find that unique solution $u(x,t)$ to (2.4).

Before finishing this lesson, we will discuss some variations of this basic problem. We start with a few modifications of the heat equation $u_t = \alpha^2 u_{xx}$.

More Diffusion-Type Equations

Lateral Heat Loss Proportional to the Temperature Difference

The equation

$$u_t = \alpha^2 u_{xx} - \beta(u - u_0) \qquad \beta > 0$$

describes heat flow in the rod with both diffusion $\alpha^2 u_{xx}$ along the rod and heat loss (or gain) across the *lateral* sides of the rod. Heat loss ($u > u_0$) or gain ($u < u_0$) is proportional to the difference between the temperature $u(x,t)$ of the rod and the surrounding medium u_0 (with β the proportionality constant). If β is very large in contrast to α^2, then the flow of heat *back and forth* along the rod will be small in contrast to the flow *in and out the sides*, and, hence, the heat will drain out the sides (at each point) according to the approximate equation $u_t = -\beta(u - u_0)$.

In chemistry where u may stand for concentration, the equation

$$u_t = \alpha^2 u_{xx} - \beta(u - u_0)$$

says that the rate of change (u_t) of the substance is due both to the diffusion $\alpha^2 u_{xx}$ (in the x-direction) and to the fact that the substance is being created ($u < u_0$) or destroyed ($u > u_0$) by a chemical reaction proportional to the difference between two concentrations u and u_0.

Internal Heat Source

The *nonhomogeneous* equation

$$u_t = \alpha^2 u_{xx} + f(x,t)$$

corresponds to the situation where the rod is being supplied with an internal heat source (everywhere along the rod and for all time t). It may be that a wire carrying electrical current passes through the rod and the resistance generates a constant heat source $f(x, t) = K$.

Diffusion-convection Equation

Suppose a pollutant is being carried along in a stream moving with velocity v. It is obvious that the concentration $u(x,t)$ of the substance changes as a function of both x (positive x measures the distance downstream) and time t. The rate of change u_t is measured by the *diffusion-convection equation*

$$u_t = \alpha^2 u_{xx} - v u_x$$

The term $\alpha^2 u_{xx}$ is the diffusion contribution and $-vu_x$ is the convection component. Whether the pollutant primarily diffuses or convects depends on the relative size of the two coefficients α^2 and v. You have probably seen smoke rising from a smoke stack. Here, the smoke particles are *convected* upward with the hot air and, at the same time, *diffuse* within the air currents.

In addition to these modifications in the heat equation, the *boundary conditions* of the rod can also be changed to correspond to other physical situations. We will discuss some of these modifications in Lesson 3.

NOTES

The heat equation $u_t = \alpha^2(x)u_{xx}$ with a variable coefficient $\alpha(x)$ would correspond to a problem where the diffusion within the rod depends on x (the material is *nonhomogeneous*). For example, if copper and steel slabs were placed next to each other (see Figure 2.3) and if the left side of the copper slabs were fixed at

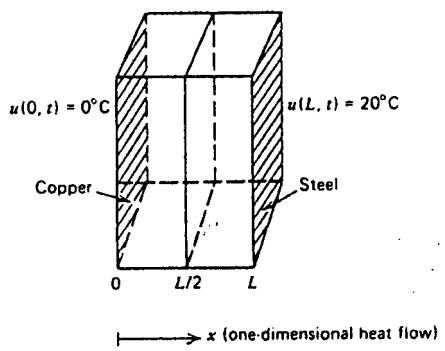

FIGURE 2.3

$u(0,t) = 0°C$ and the right side of the steel sheet were fixed at $u(L,t) = 20°C$, then the PDE that describes the heat flow would be

$$u_t = \alpha^2(x)u_{xx} \qquad 0 < x < L$$

$$\text{where } \alpha(x) = \begin{cases} \alpha_1 \text{ (diffusion coefficient of copper)} & 0 < x < L/2 \\ \alpha_2 \text{ (diffusion coefficient of steel)} & L/2 < x < L \end{cases}$$

PROBLEMS

1. If the initial temperature of the rod were

$$u(x,0) = \sin \pi x \qquad 0 \leq x \leq 1$$

and if the BCs were

$$u(0,t) = 0$$
$$u(1,t) = 0$$

what would be the behavior of the rod temperature $u(x,t)$ for later values of time?

HINT Use the physical interpretation of the heat equation $u_t = \alpha^2 u_{xx}$.

2. Suppose the rod has a constant internal heat source, so that the basic equation describing the heat flow within the rod is

$$u_t = \alpha^2 u_{xx} + 1 \qquad 0 < x < 1$$

Suppose we fix the boundaries' temperatures by $u(0,t) = 0$ and $u(1,t) = 1$. What is the steady-state temperature of the rod? In other words, does the temperature $u(x,t)$ converge to a constant temperature $U(x)$ independent of time?

HINT Set $u_t = 0$. It would be useful to graph this temperature. Also start with an initial temperature of zero and draw some temperature profiles.

3. Suppose a metal rod loses heat across the lateral boundary according to the equation

$$u_t = \alpha^2 u_{xx} - \beta u \qquad 0 < x < 1$$

and suppose we keep the ends of the rod at $u(0,t) = 1$ and $u(1,t) = 1$. Find the steady-state temperature of the rod (graph it). Where is heat flowing in this problem?

4. Suppose a laterally insulated metal rod of length $L = 1$ has an initial temperature of $\sin(3\pi x)$ and has its left and right ends fixed at temperatures zero and 10°C. What would be the IBVP that describes this problem?*

*Note that the boundary and initial data do not match up in this problem.

OTHER READING

1. *Equations of Mathematical Physics* by A. N. Tikhonov and A. A. Samarskii. Macmillan, 1963; Dover, 1990. An encyclopedia of information; contains many good examples and problems.

LESSON 3

Boundary Conditions for Diffusion-Type Problems

PURPOSE OF LESSON: To show how heat-flow and diffusion-type problems can give rise to a variety of boundary conditions and to introduce the important concept of *flux*.

Three important types of BCs discussed are

1. $u = g(t)$ (temperature specified on the boundary).
2. $\dfrac{\partial u}{\partial n} + \lambda u = g(t)$ (temperature of the *surrounding medium* is specified; n is the *outward normal* direction to the boundary).
3. $\dfrac{\partial u}{\partial n} = g(t)$ (heat flow across the boundary specified).

When describing the various types of boundary conditions that can occur for heat-flow problems, three basic types generally come to mind. Lesson 3 discusses these three kinds of BCs and gives an example of how they occur in experiments.

Type 1 BC (Temperature specified on the boundary)

Consider the heat flow in the one-dimensional rod illustrated in Figure 3.1 and suppose we make the ends of the rod follow the temperature curves $g_1(t)$ and $g_2(t)$.

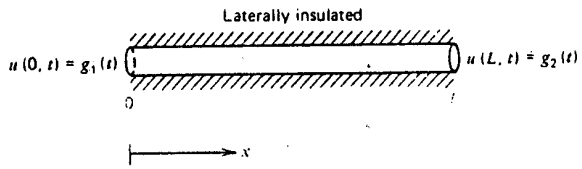

FIGURE 3.1 Temperature specified on the boundary.

As we mentioned in the previous lesson, an apparatus that keeps the ends at specified temperatures requires a thermostat at each end and heating elements to adjust the temperature accordingly. Problems with BCs of this kind are fairly common. It may even be that the goal of the problem is to find the *boundary temperatures* (boundary control) $g_1(t)$ and $g_2(t)$ that will force the temperature to behave in a suitable manner. In the steel industry, it is often necessary to determine the *boundary controls* so that the temperature of the metal inside the furnace changes over time but the temperature *gradient* from one point to another is small.

Similar types of BCs also apply to higher dimensional domains, for example, in two dimensions, we could imagine the interesting problem of finding the temperature inside the circular disc (of radius R) when the boundary temperature is specified in polar coordinates to be

$$u(R,\theta,t) = \cos t \sin \theta$$

See Figure 3.2.

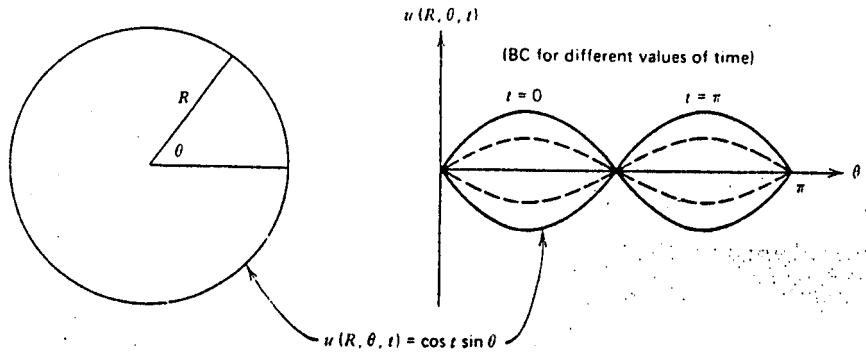

FIGURE 3.2 Oscillating boundary temperature.

Of course, we'd have to have an initial temperature to get this experiment started, but in this case, the effects of our IC would vanish after a short period of time, and the resulting temperature inside the circle would depend on the boundary temperature.

Type 2 BC (Temperature of the surrounding medium specified)

Suppose we consider again our laterally insulated copper rod, but now instead of requiring the two boundaries to be specified at temperatures $g_1(t)$ and $g_2(t)$, we only bring them in contact with surrounding mediums that have those temperatures. In other words, suppose the left side of the rod is enclosed in a

container of liquid that has a changing temperature $g_1(t)$, while the right end is enclosed in another liquid with temperature $g_2(t)$ (Figure 3.3).

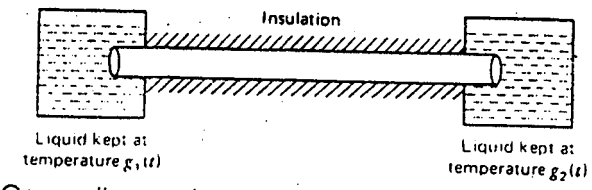

FIGURE 3.3 Convection cooling at the boundaries.

By specifying these types of BCs, we cannot say the boundary temperatures of the rod will be the same as the liquid temperatures $g_1(t)$ and $g_2(t)$, but we do know (Newton's law of cooling) that whenever the rod temperature at one of the boundaries is *less* than the respective liquid temperatures, then heat will flow into the rod at a rate proportional to this difference. In other words, for the one-dimensional rod with boundaries at $x = 0$ and L, Newton's law of cooling states

$$(3.1)\qquad \begin{cases} \text{Outward flux of heat (at } x = 0) = h[u(0,t) - g_1(t)] \\ \text{Outward flux of heat (at } x = L) = h[u(L,t) - g_2(t)] \end{cases}$$

where h is a **heat-exchange coefficient,** which is a measure of how many calories flow across the boundary per unit of temperature difference per second per cm and the **outward flux of heat** is the number of calories crossing the ends of the rod per second. Note that the outward flux of heat will be positive at either end provided the temperature of the rod is greater than the surrounding medium. Equations (3.1) can now be used in conjunction with what is known as Fourier's Law of Cooling to arrive at our BCs. Fourier's law gives us another representation (the first one is 3.1) for the outward flux of heat and by setting these two representations equal to each other, we get our BC's. First, we state Fourier's law (proven experimentally):

(3.2) | Outward flux of heat across a boundary is proportional to the inward normal derivative across the boundary. |

This law says that if the temperature is increasing rapidly in the direction *outward* from the boundary of D (Figure 3.4), then heat will flow *from* the surrounding medium *into* the domain D.

In our one-dimensional problem, Fourier's law takes the form:

$$(3.3)\qquad \begin{cases} \text{Outward flux of heat (at } x = 0) = k\dfrac{\partial u(0,t)}{\partial x} \\ \text{Outward flux of heat (at } x = L) = -k\dfrac{\partial u(L,t)}{\partial x} \end{cases}$$

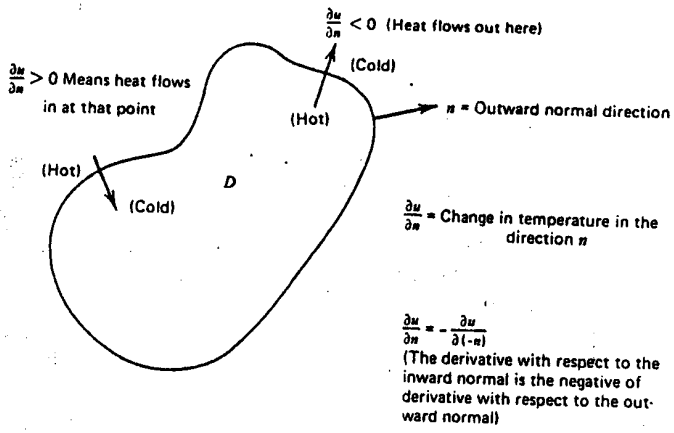

FIGURE 3.4 Illustration of Fourier's law.

where k is the **thermal conductivity** of the metal, which is a measure of how well the material conducts heat. (Poorly conducting materials have values near zero in cgs. units, while copper and aluminum have values close to one.)

Fourier's law (3.3) actually holds anywhere inside the rod and not just at the boundary; for example,

$$\text{Flux of heat crossing } x_0 \text{ (from left to right)} = -kA\frac{\partial u}{\partial x}(x_0,t) \tag{3.4}$$

See Figure 3.5.

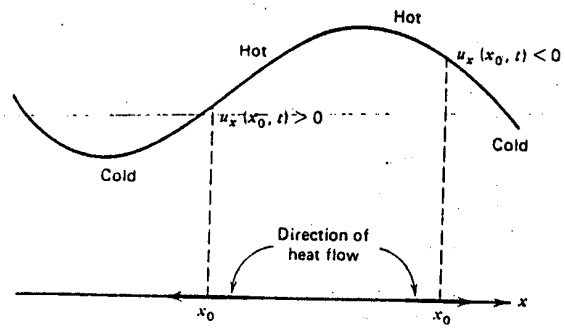

FIGURE 3.5 Another illustration of Fourier's law.

Fourier's law (3.4) says that if $u_x(x_0,t) < 0$, then heat will flow from *left to right*; if $u_x(x_0,t) > 0$, then the flow of heat through point x_0 will be from *right to left* (heat always flows from high to low temperatures).

Finally, if we use the two expressions (3.1) and (3.3) for heat flux, we have our desired BCs for Figure 3.3 in purely mathematical terms; namely,

$$\text{BCs} \quad \begin{cases} \dfrac{\partial u(0,t)}{\partial x} = \dfrac{h}{k}\left[u(0,t) - g_1(t)\right] \\[2ex] \dfrac{\partial u(L,t)}{\partial x} = -\dfrac{h}{k}\left[u(L,t) - g_2(t)\right] \end{cases} \qquad 0 < t < \infty$$

Quite often, the constant h/k is simply written as λ, and so we have the BCs for heat flow across the boundary

$$\begin{aligned} u_x(0,t) &= \lambda\left[u(0,t) - g_1(t)\right] \\ u_x(L,t) &= -\lambda\left[u(L,t) - g_2(t)\right] \end{aligned} \tag{3.5}$$

In higher dimensions, we have similar BCs; for example, if the boundary of a circular disc is interfaced with a moving liquid that has a temperature $g(\theta,t)$, our BC would be

$$\frac{\partial u}{\partial r}(R,\theta,t) = -\frac{h}{k}\left[u(R,\theta,t) - g(\theta,t)\right]$$

Here, $\frac{\partial u}{\partial r}(R,\theta,t)$ represents the outward normal derivative (in the positive r-direction) of u evaluated at a point (R,θ) on the boundary. This type of BC would be called a *linear* BC (since it is linear in u and u_r) but nonhomogeneous due to the right-hand side $g(\theta,t)$.

Type 3 BC (Flux specified—including the special case of insulated boundaries)

Insulated boundaries are those that do not allow any flow of heat to pass, and, hence, the normal derivative (inward or outward) must be zero on the boundary (since the normal derivative is proportional to the flux). In the case of the one-dimensional rod with insulated ends at $x = 0$ and $x = L$, the BCs are

$$\begin{aligned} u_x(0,t) &= 0 \\ u_x(L,t) &= 0 \end{aligned} \qquad 0 < t < \infty$$

In two-dimensional domains, an insulated boundary would mean that the *normal derivative* of the temperature across the boundary is zero. For example, if the circular disc were insulated on the boundary, then the BC would be $u_r(R,\theta,t) = 0$ for all $0 \leq \theta < 2\pi$ and all $0 < t < \infty$.

On the other hand, if we specify the amount of heat entering across the boundary of our disc, the BC is

$$u_r(R,\theta,t) = f(\theta,t)$$

where $f(\theta,t)$ would represent the amount of heat crossing *into* the circular disc from an outside heating source.

We now illustrate different types of BCs.

Typical BCs for One-Dimensional Heat Flow

Suppose we have a copper rod 200 cm long that is laterally insulated and has an initial temperature of 0°C. Suppose the top of the rod ($x = 0$) is insulated, while the bottom ($x = 200$) is immersed in moving water that has a constant temperature of $g_2(t) = 20$°C (Figure 3.6).

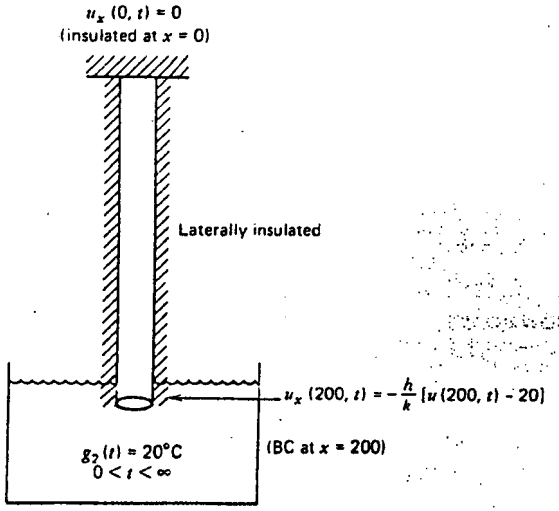

FIGURE 3.6 Initial-boundary-value problem.

The mathematical model for this problem would be the following four equations:

$$\text{PDE} \qquad u_t = \alpha^2 u_{xx} \qquad 0 < x < 200 \qquad 0 < t < \infty$$

$$(3.6) \qquad \text{BCs} \qquad \begin{cases} u_x(0,t) = 0 \\ u_x(200,t) = -\dfrac{h}{k}[u(200,t) - 20] \end{cases} \qquad 0 < t < \infty$$

$$\text{IC} \qquad u(x,0) = 0°\text{C} \qquad 0 \leq x \leq 200$$

where

α^2 = 1.16 cm²/sec (diffusivity constant for copper)

k = 0.93 cal/cm-sec°C (thermal conductivity of copper)

h = heat exchange coefficient. To find h is a hard problem in itself. It measures the rate that heat is being exchanged between the bottom of the rod and the surrounding water. It is a function of how fast the water is being circulated, the nature of the interface, and so forth. The reader would have to carry out an experiment to determine its value.

NOTES

1. A typical heat-flow problem inside a square is shown in Figure 3.7.

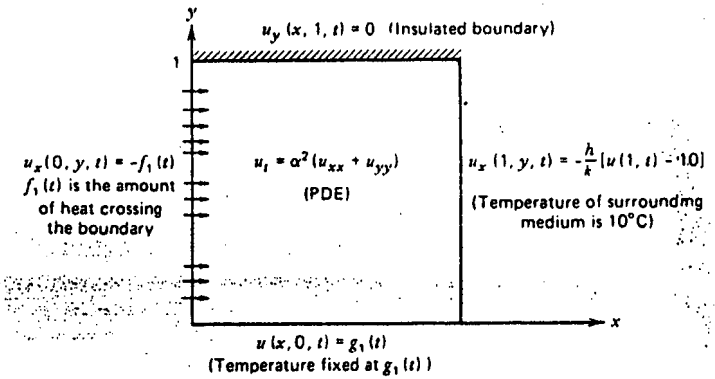

FIGURE 3.7 Typical BCs for diffusion problems inside a square.

In this problem, after we specify the initial temperature $u(x,y,0)$ at $t = 0$ inside the square, the PDE and BCs in the diagram will take over for $0 < t < \infty$ and determine the subsequent temperature values $u(x,y,t)$. Whatever the temperature is, however, it must satisfy the BCs in Figure 3.7.

2. Note that the BC

$$u_r(R,\theta,t) = -\frac{h}{k}[u(R,\theta,t) - g(\theta,t)]$$

on the circle will not require the boundary temperature to be $g(\theta,t)$, but when the heat-exchange coefficient h is large, then the BC essentially says that the boundary temperature $u(R,\theta,t)$ is almost equal to $g(\theta,t)$.

PROBLEMS

1. Draw rough sketches of the solution to the IBVP (3.6) for different values of time. Do your sketches satisfy the BCs? What is the steady-state temperature of the rod? Is this obvious based on your intuition?
2. What is your interpretation of the initial-boundary-value problem?

$$\text{PDE} \qquad u_t = \alpha^2 u_{xx} \qquad 0 < x < 1 \qquad 0 < t < \infty$$

$$\text{BCs} \qquad \begin{cases} u(0,t) = 0 \\ u_x(1,t) = 1 \end{cases} \qquad 0 < t < \infty$$

$$\text{IC} \qquad u(x,0) = \sin(\pi x) \qquad 0 \leq x \leq 1$$

Can you draw rough sketches of the solution for different values of time? Will the solution come to a steady state; is this obvious?

3. What is your physical interpretation of the problem?

$$\text{PDE} \qquad u_t = \alpha^2 u_{xx} \qquad 0 < x < 1 \qquad 0 < t < \infty$$

$$\text{BCs} \qquad \begin{cases} u_x(0,t) = 0 \\ u_x(1,t) = 0 \end{cases} \qquad 0 < t < \infty$$

$$\text{IC} \qquad u(x,0) = \sin(\pi x) \qquad 0 \leq x \leq 1$$

Can you draw rough sketches of this solution for various values of time? What about the steady-state temperature?

4. Suppose a metal rod laterally insulated has an initial temperature of 20°C but immediately thereafter has one end fixed at 50°C. The rest of the rod is immersed in a liquid solution of temperature 30°C. What would be the IBVP that describes this problem?

OTHER READING

1. *Conduction of Heat in Solids* by H. S. Carslaw and J. C. Jaeger. Oxford University Press, 1959. An excellent reference that discusses BCs of many physical problems.
2. *Partial Differential Equations in Biology* by C. S. Peskin. Courant Institute of Mathematical Sciences, 1976. Several biological phenomena such as nerve cells, the inner ear, and the cardiovascular system are modeled by PDEs.

LESSON 4

Derivation of the Heat Equation

PURPOSE OF LESSON: To show how the one-dimensional heat equation

$$u_t = \alpha^2 u_{xx} + f(x,t)$$

is derived from the basic principle of *conservation of heat*. Physical concepts such as *thermal conductivity*, *thermal capacity*, and *density* are discussed, and it is shown how the rate of heat transfer depends on these three basic physical parameters. A few variations of the basic heat equation are also discussed.

In all areas of science, we begin with a given set of assumptions that are taken to be self-evident and from which all other ideas are derived. Of course, what is self-evident to one person may hold doubts for others. The history of science consists of pushing back the basic axioms further and further so that there is a universally agreed upon starting point.

For example, one person may think that all relevant facts will spring from a basic assumption, say assumption B. From assumption B, he or she may prove theorem C, which in turn proves theorem D, which in turn proves others (Figure 4.1).

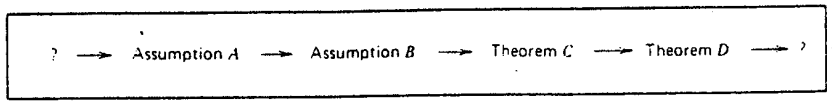

FIGURE 4.1 The axiomatic method.

This of course is progress—the more new results a person can prove, the better. Physicists, chemists, and biologists all proceed in this basic manner.

On the other hand, instead of proving new theorems we may ask if it is possible to find a new assumption, say assumption A more basic than assumption B, so that assumption B can be proven from A. In this way, we are pushing back the frontiers of knowledge. In the general area of heat flow, the concept of *conservation of energy* (heat energy) is the basis from which other principles are derived (Figure 4.2).

Conservation of energy (Assumption)	$\longrightarrow$	$u_t = \alpha^2 u_{xx} + f(x,t)$	$\longrightarrow$	Other properties of heat flow

FIGURE 4.2 Conservation of energy: the cornerstone of heat-flow problems.

We could, of course, forget this lesson and use the heat equation as the starting point (some people may think it is self-evident in itself), but this would be shortchanging serious students, since conservation of energy assumptions are basic to science. Scientists often begin modeling specific problems by writing conservation of energy relationships and then rewriting them as partial differential equations.

We now turn to the goal of the lesson—to derive the heat equation from the conservation of heat equation.

Derivation of the Heat Equation

Suppose we have a one-dimensional rod of length L for which we make the following assumptions:

1. The rod is made of a single homogeneous conducting material.
2. The rod is laterally insulated (heat flows only in the x-direction).
3. The rod is thin (the temperature at all points of a cross section is constant).

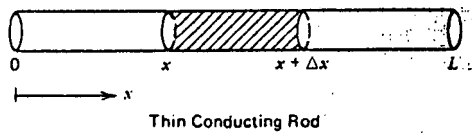

FIGURE 4.3 Thin conducting rod.

If we apply the principle of conservation of heat to the segment $[x, x + \Delta x]$, we can claim

$$\text{(4.1)} \qquad \begin{aligned} &\text{Net change of heat inside } [x, x + \Delta x] \\ &= \text{Net flux of heat across the boundaries} \\ &+ \text{Total heat generated inside } [x, x + \Delta x] \end{aligned}$$

Now, inasmuch as the total amount of heat (in calories) inside $[x, x + \Delta x]$ at any time t is measured by (see reference 1 in this lesson)

$$\text{Total heat inside } [x, x + \Delta x] = \int_x^{x+\Delta x} c\rho A u(s,t)\, ds$$

where

c = thermal capacity of the rod (measures the ability of the rod to store heat).
ρ = density of the rod.
A = cross-section area of the rod

we can write the conservation of energy equation (4.1) via calculus as

$$\text{(4.2)} \qquad \begin{aligned} \frac{d}{dt}\int_x^{x+\Delta x} c\rho A u(s,t)\, ds &= c\rho A \int_x^{x+\Delta x} u_t(s,t)\, ds \\ &= kA\,[u_x(x + \Delta x, t) - u_x(x,t)] + A\int_x^{x+\Delta x} f(s,t)\, ds \end{aligned}$$

where

k = thermal conductivity of the rod (measures the ability to conduct heat).
$f(x,t)$ = external heat source (calories per cm per sec).

The problem now is to replace equation (4.2) by one that does not contain integrals. The reader may recall the Mean Value Theorem from calculus.

Mean Value Theorem

If $f(x)$ is a continuous function on $[a,b]$, then there exists at least one number ξ, $a < \xi < b$ that satisfies

$$\int_a^b f(x)dx = f(\xi)\,(b - a)$$

Applying this result to equation (4.2), we arrive at the following equation:

$$c\rho A u_t(\xi_1, t)\Delta x = kA[u_x(x + \Delta x, t) - u_x(x,t)] + Af(\xi_2, t)\,\Delta x \quad x < \xi < x + \Delta x$$

or

$$u_t(\xi, t) = \frac{k}{c\rho}\left\{\frac{u_x(x + \Delta x, t) - u_x(x,t)}{\Delta x}\right\} + \frac{1}{c\rho}f(\xi, t)$$

Finally letting $\Delta x \to 0$, we have the desired result

$$\text{(4.3)} \qquad \boxed{u_t(x,t) = \alpha^2 u_{xx}(x,t) + F(x,t)}$$

where